LAB:5

TASK 1:
Print table of any number

```
#include <iostream>
using namespace std;
int main ( )
    {
  int num;
      cout << "Enter number:";
      cin >> num;
        for (int i=1; i<=10; ++){
      cout << num<< " * "<< i<< " = "<< num* i<<endl
                  }
       return 0;
            }
```

output:

```
Enter number:2
2*1 = 2
2*2 = 4
2*3 = 6
2*4 = 8
2*5 = 10
2*6 = 12
2*7 = 14
2*8 = 16
2*9 = 18
2*10 = 20
```

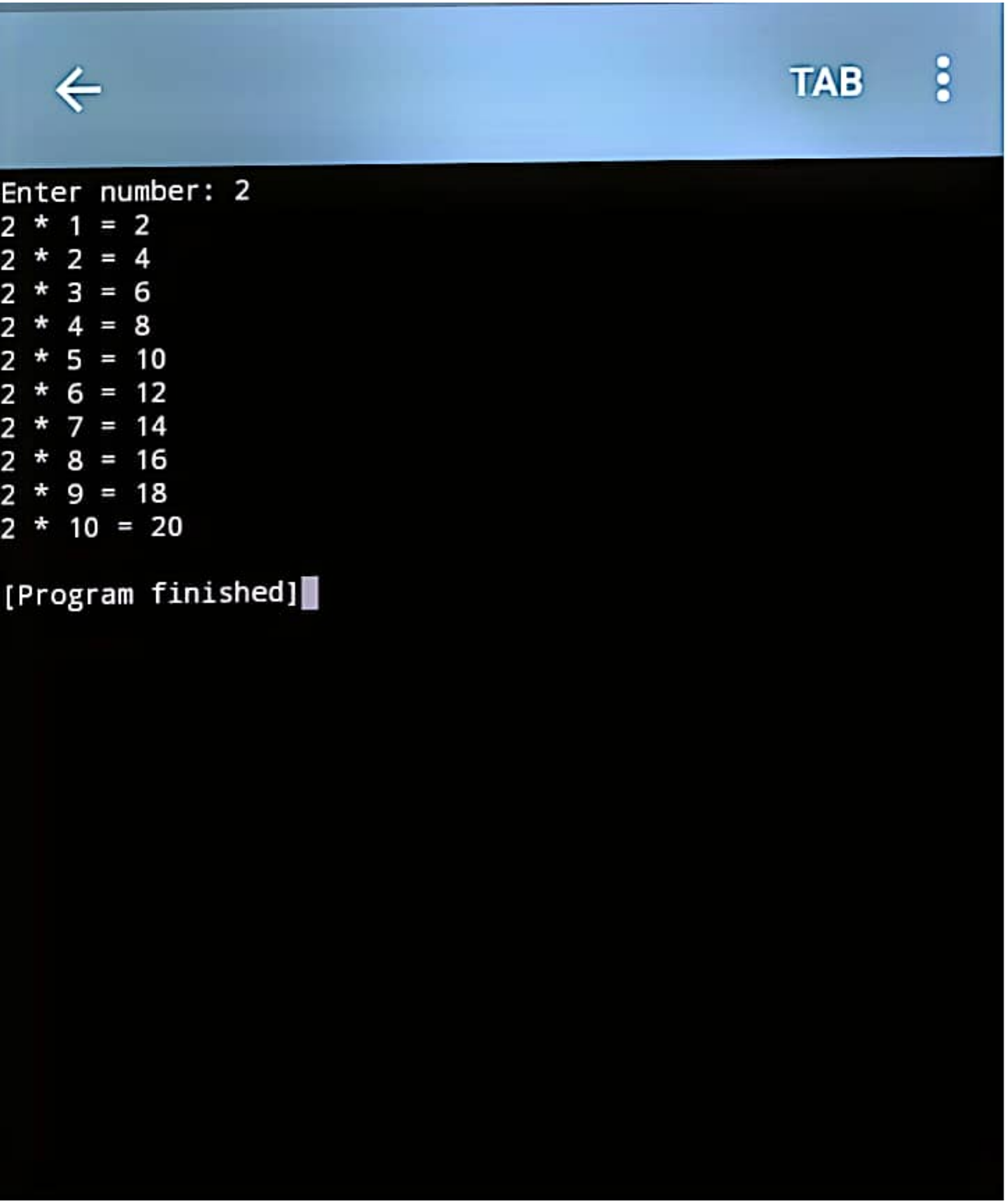
Scanned with CamScanner

new*

```
1  #include <iostream>
2  using namespace std;
3  int main() {
4      int num;
5      cout << "Enter number: ";
6      cin >> num;
7      for(int i=1; i<=10; i++){
8          cout << num << " * " << i << " = " <<
   num*i << endl;
9      }
10     return 0;
11 }
```

Tab | { | } | : | ; | "

Scanned with CamScanner



Scanned with CamScanner

TASK 2:

find power of any number using for loop

```
#include <iostream>
using namespace std;
int main () {
    int base, exp;
    long long power = 1;
    cout << "Enter base and Exponent: ";
    cin >> Base >> exp;
    for(int i=1 ; i<=exp; i++) {
      power = power * Base; }
    cout << base << "^" << exp << "=" << power;
    return 0;
}
```

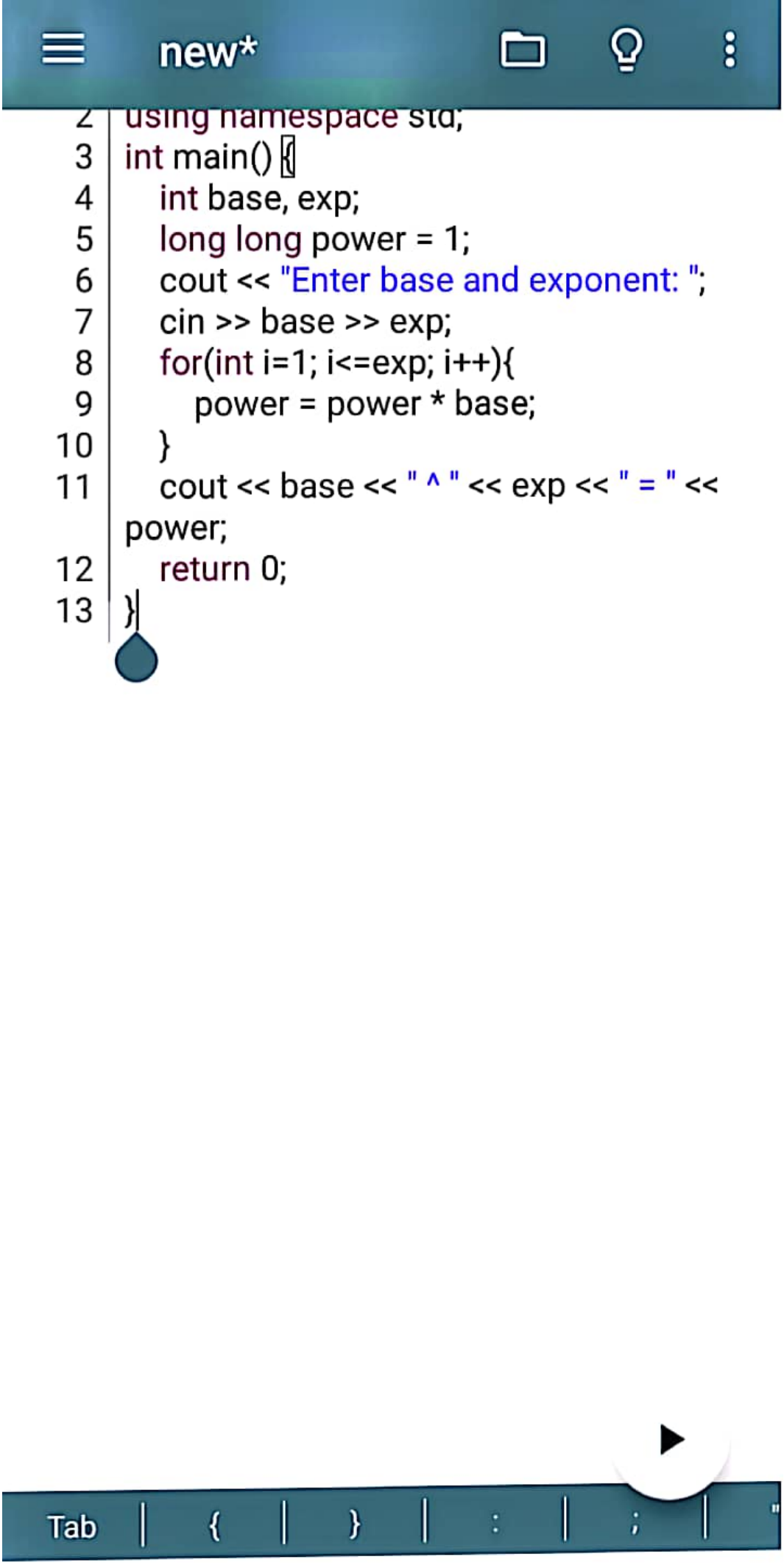
output:

Enter base and Exponent :34

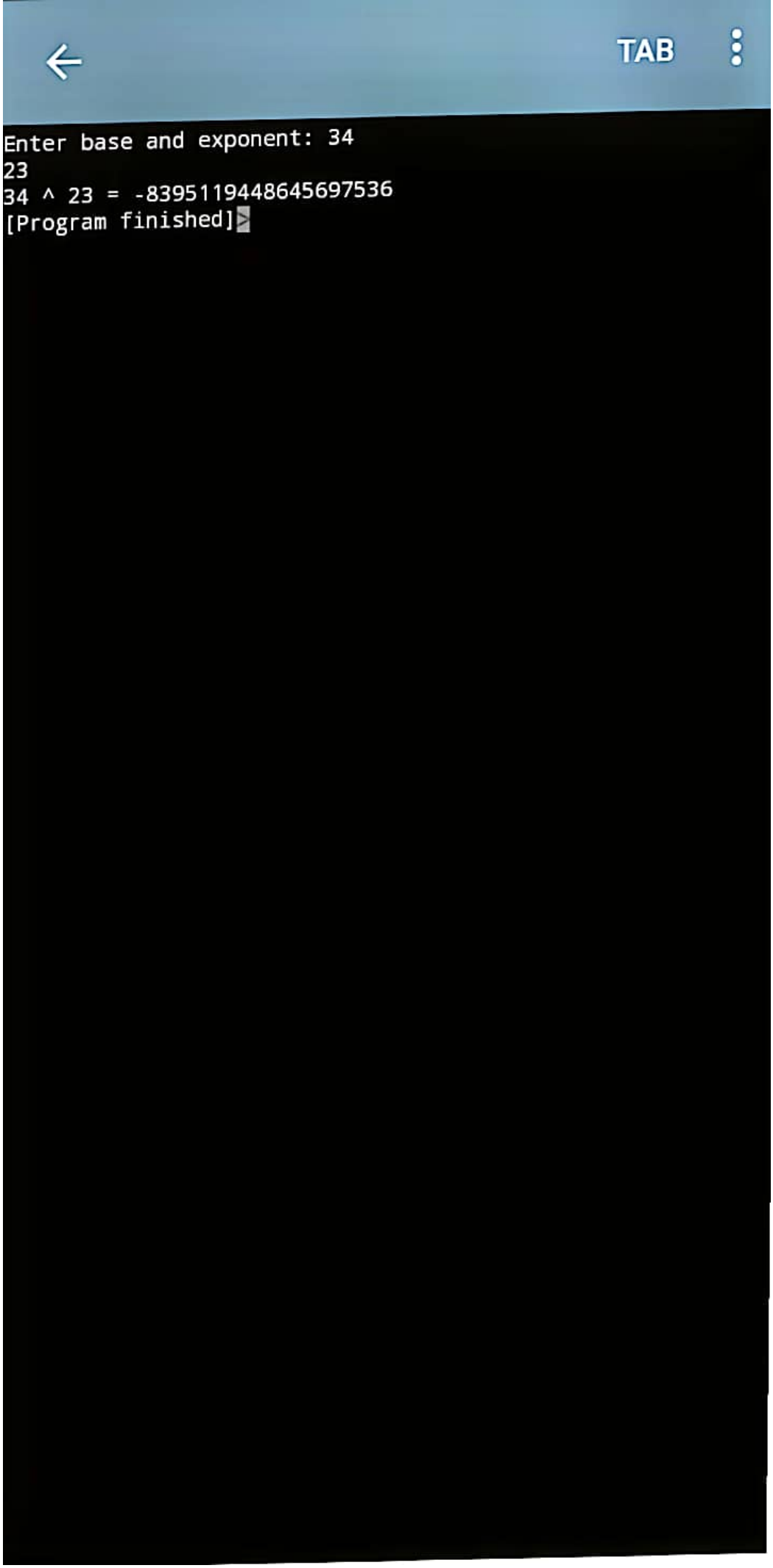
23

34 ^ 23 = -8395119448645647538

Scanned with CamScanner



Scanned with CamScanner



Scanned with CamScanner

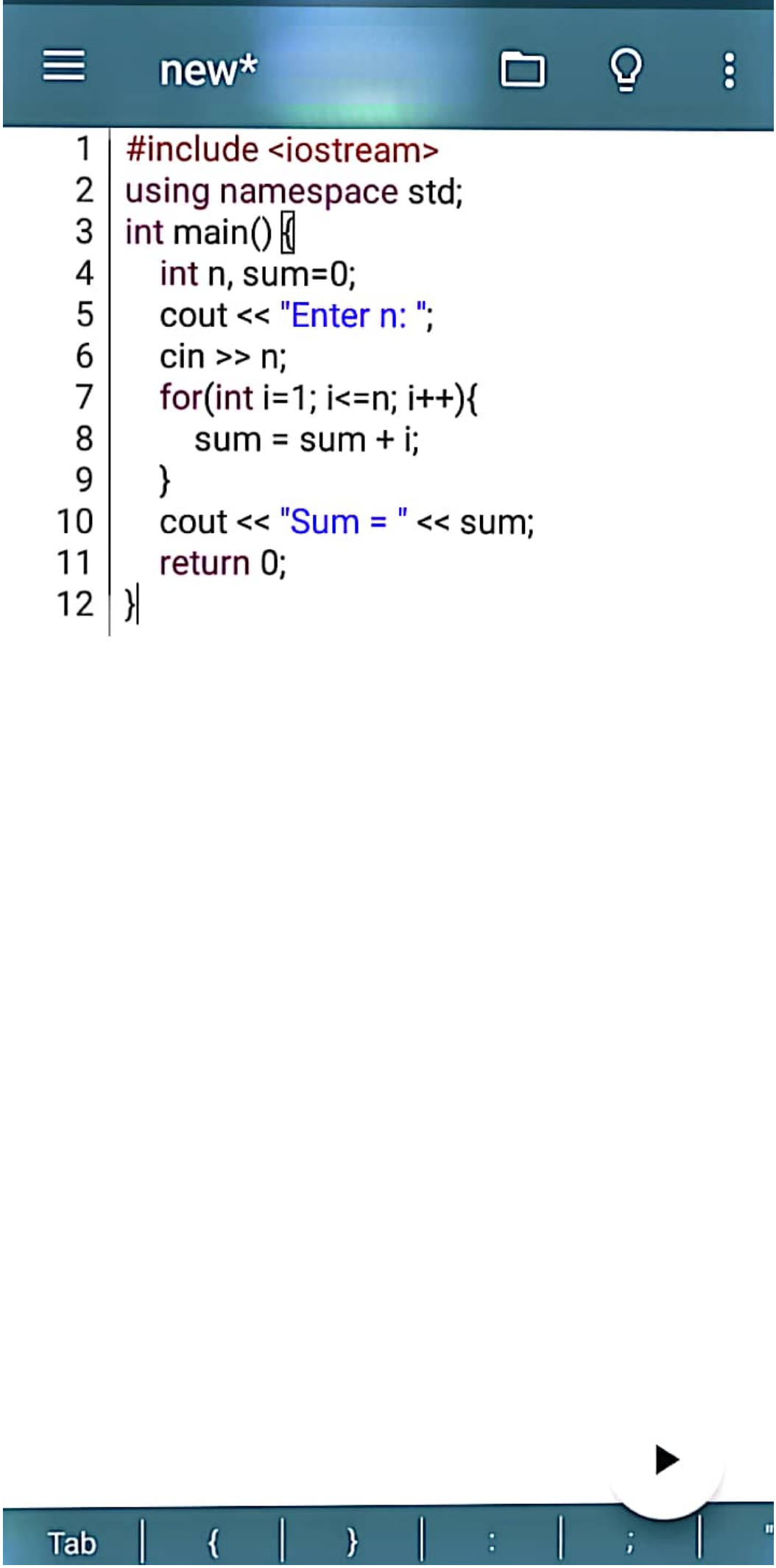
TASK 3:
Sum of Natural Numbers

```
#include <iostream>
using namespace std;
int main () {
    int n, sum=0
    cout <<" Enter n: ";
    cin>> n;
    for (int i=1; i<=n; i++) {
        sum=sum+i; }
    cout <<"sum =" <<sum;
    return 0;
}
```

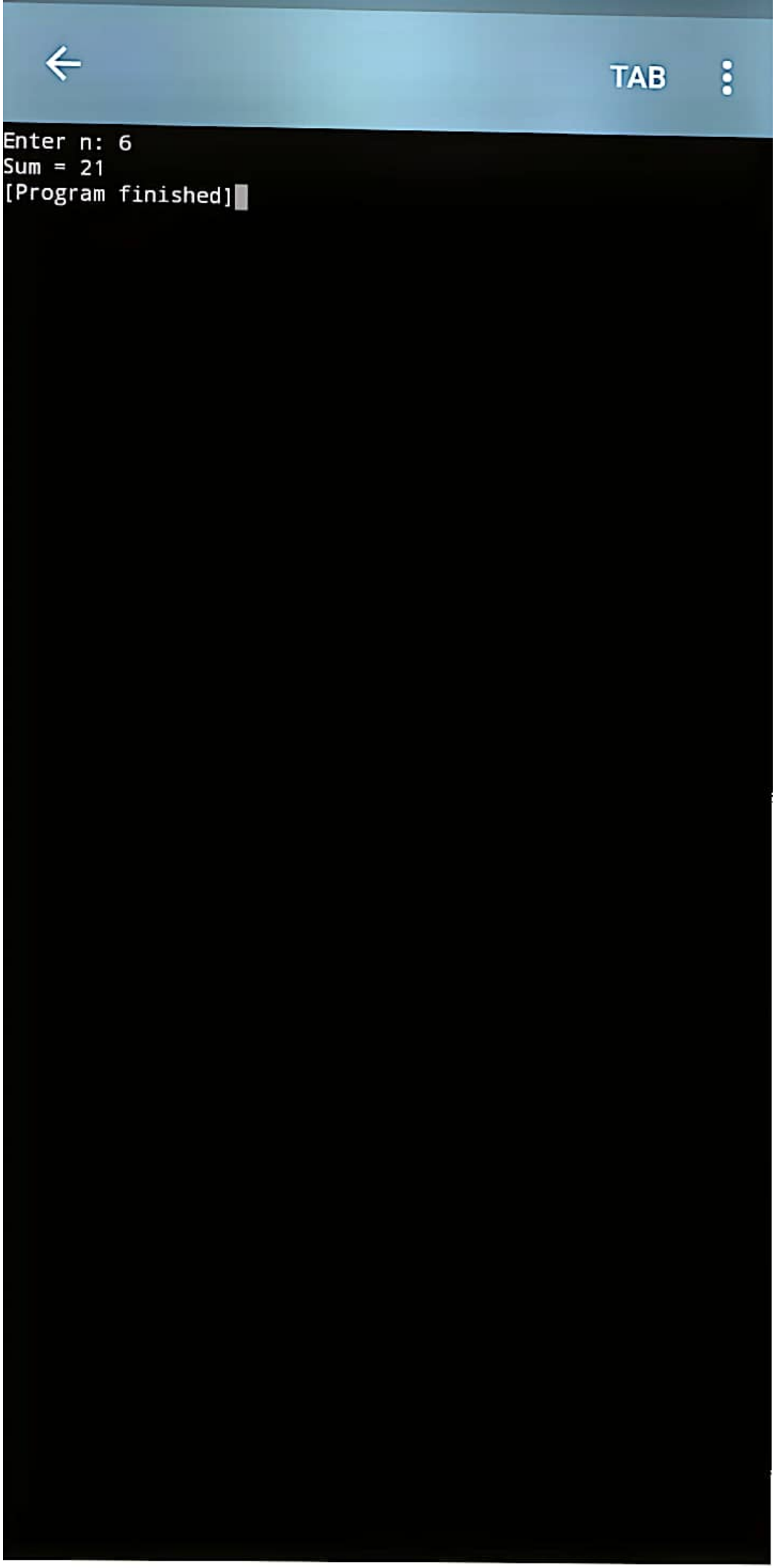
output:

```
Enter n : 6
Sum = 21
```

Scanned with CamScanner



Scanned with CamScanner



Scanned with CamScanner

Task 4:

sum of Digits:

```
#include <iostream>
using namespace std;
int main () {
    int num, sum=0, rem;
    cout << "Enter number: ";
    cin >> num;
    while (num >0) {
        rem = num%10;
        sum = sum + rem;
        num = num /10; }
    cout <<" sum of digits = "<<sum;
    return 0;
}
```

output:

Enter number: 42

sum of digits = 6

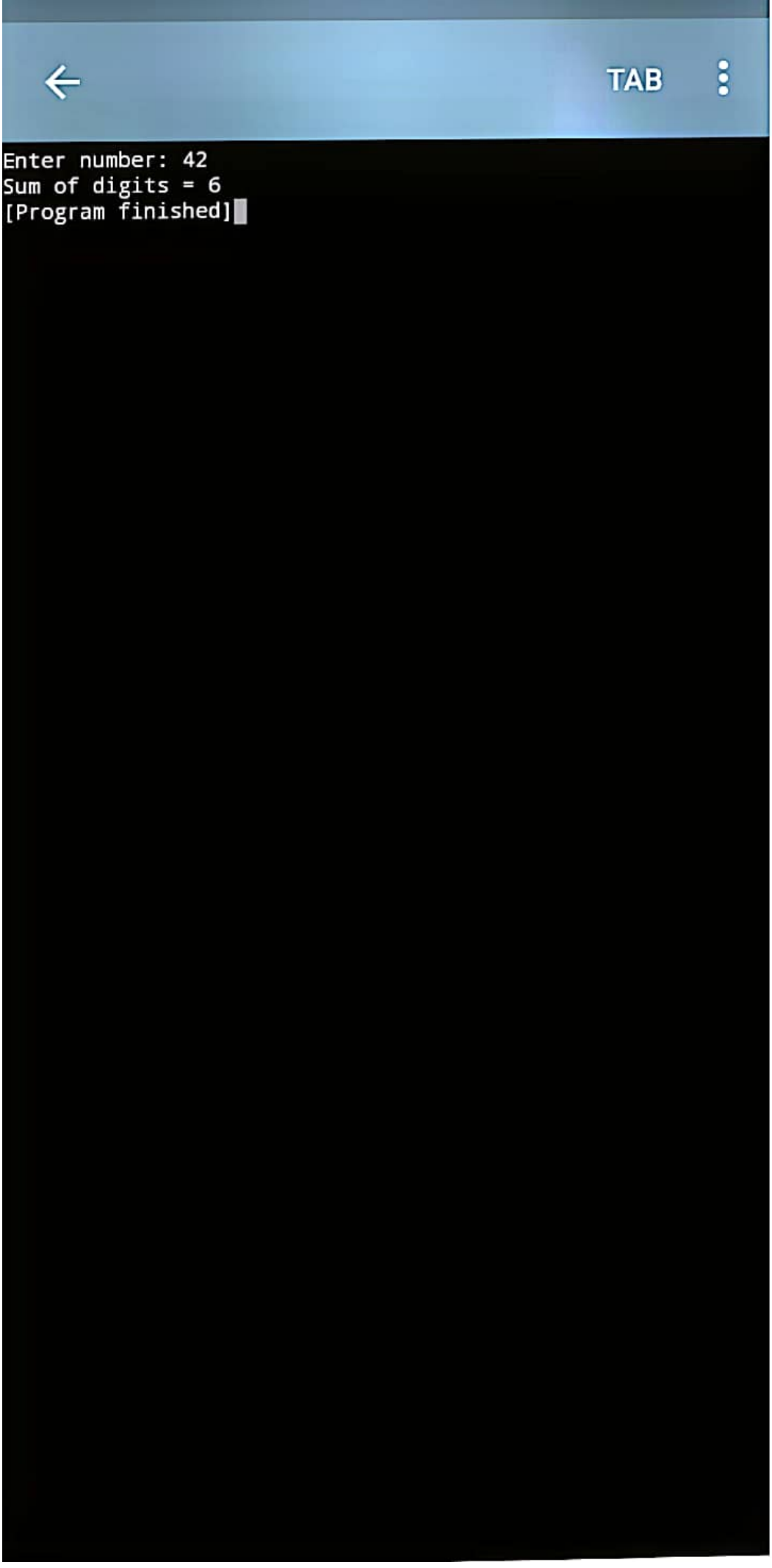
Scanned with CamScanner

new*

```
1  #include <iostream>
2  using namespace std;
3  int main() {
4      int num, sum=0, rem;
5      cout << "Enter number: ";
6      cin >> num;
7      while(num > 0){
8          rem = num % 10;
9          sum = sum + rem;
10         num = num / 10;
11     }
12     cout << "Sum of digits = " << sum;
13     return 0;
14 }
```

Tab | { | } | : | ; | "

Scanned with CamScanner



Scanned with CamScanner

TASK 5

factorial of certain number

```
#include <iostream>
using namespace std;
int main () {
    int n;
    long long fact =1;
    cout << "Enter number:";
    cin>> n;
    for (int i=1 ; i<=n ; i++) {
        fact = fact * i ; }
cout << " factorial of " << n << " = " << fact;
    return 0 ;
    }
```

output :

Enter number : 31

factorial of 31 = 4999213071378415616

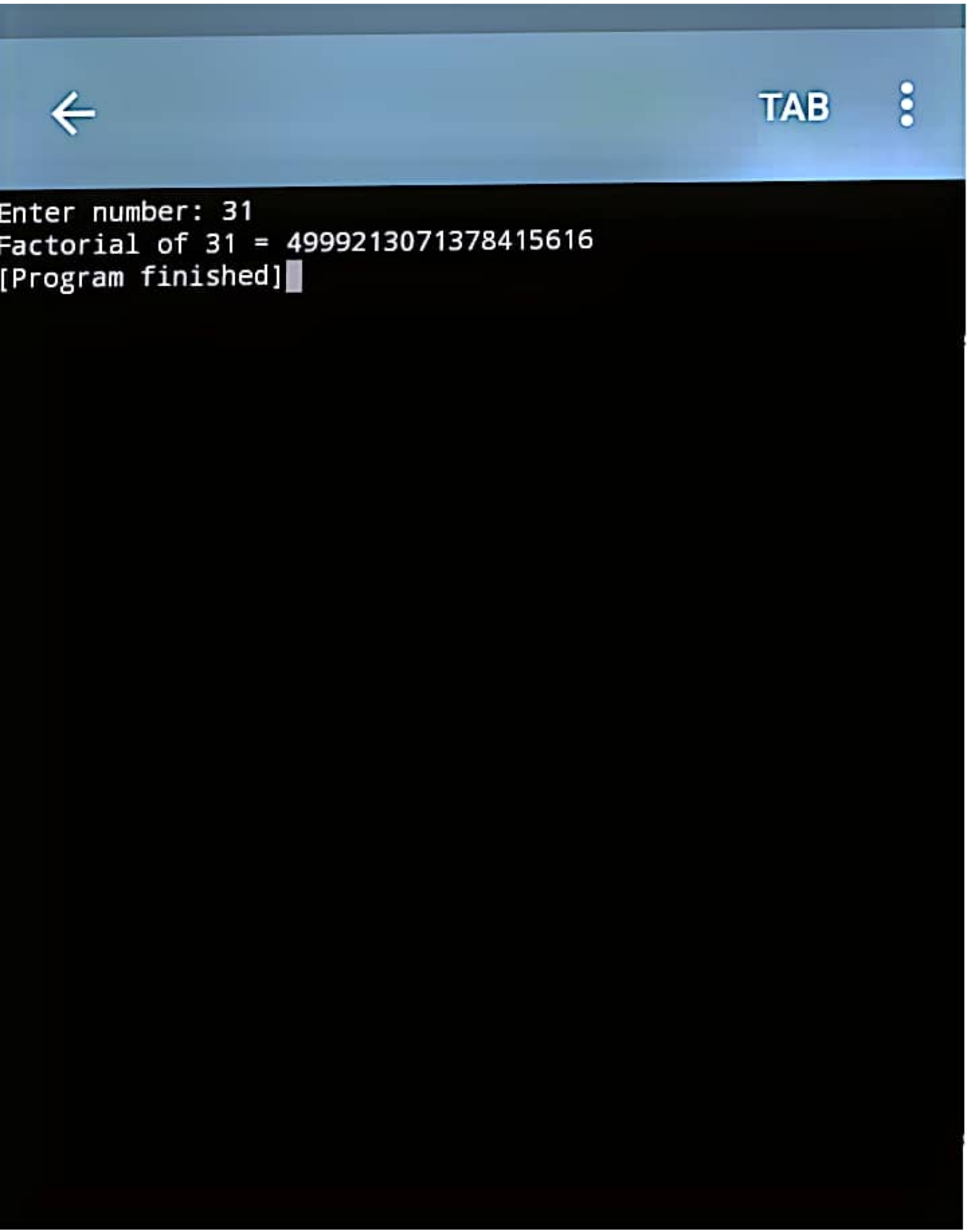
Scanned with CamScanner

new*

```
1  #include <iostream>
2  using namespace std;
3  int main() {
4      int n;
5      long long fact=1;
6      cout << "Enter number: ";
7      cin >> n;
8      for(int i=1; i<=n; i++){
9          fact = fact * i;
10     }
11     cout << "Factorial of " << n << " = " <<
   fact;
12     return 0;
13 }
14 // 4! = 4*3*2*1 = 24
```

Tab | { | } | : | ; | "

Scanned with CamScanner



Scanned with CamScanner

TASK 6:

Print Pattern:

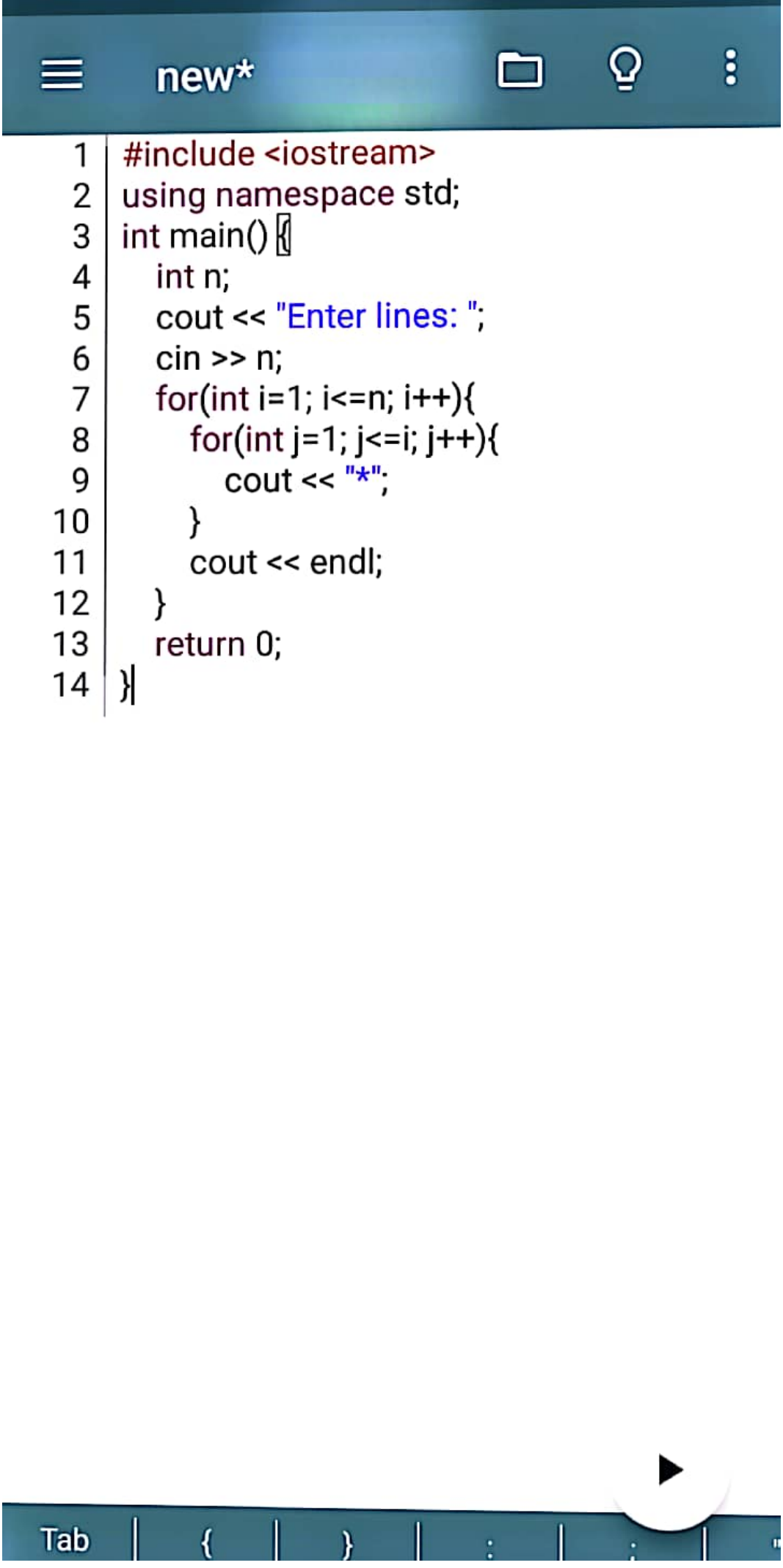
```
#include <iostream>
using namespace std;
int main () {
int n;
cout << "Enter lines: ";
cin >> n;
for (int i=1; i<n; i++) {
 for (int j=1 ; j<=i ; j++){
   cout << "* "
 }
 cout << endl; }
return 0;
}
```

output:

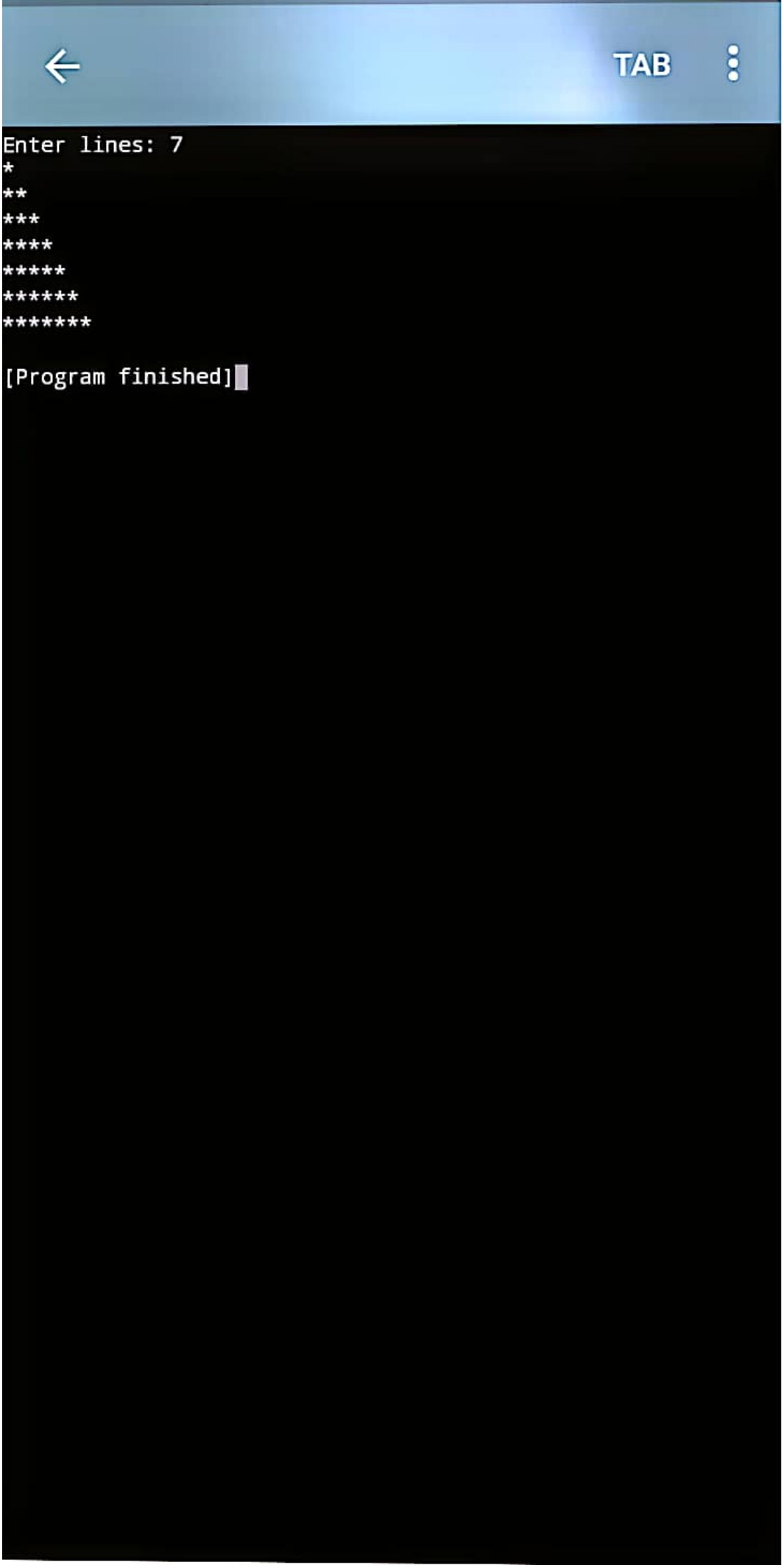
Enter lines : 7

```
*
* *
* * *
* * * *
* * * * *
* * * * * *
* * * * * * *
```

Scanned with CamScanner



Scanned with CamScanner



Scanned with CamScanner

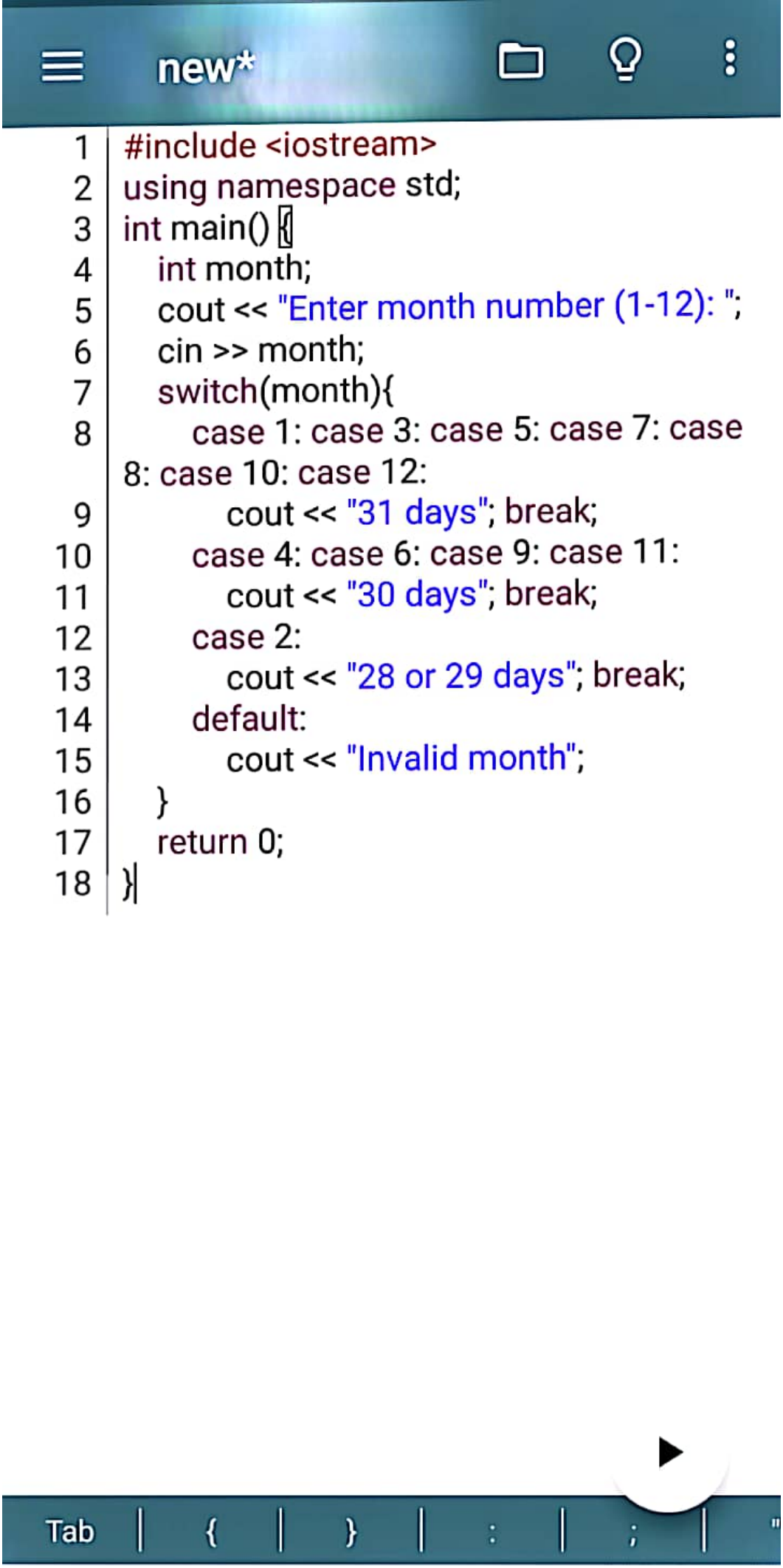
Total number of days in a month using switch case:

```
# include <iostream>
using namespace std;
int main () {
    int month;
    cout<<"Enter month number(1-12): ";
    Cin >> month;
    switch(month) {
     case 1: case 3: case 5: case 7: case 8: case 10: case 12
 cout << "30 days"; break;
 case 4: case 6: case 9 : case :
  cout << "30 days"; Break;
 case 2:
  cout<<"28 or 29 day"; break;
   default :
   cout << "invalid month"; }
     return 0;
}
```

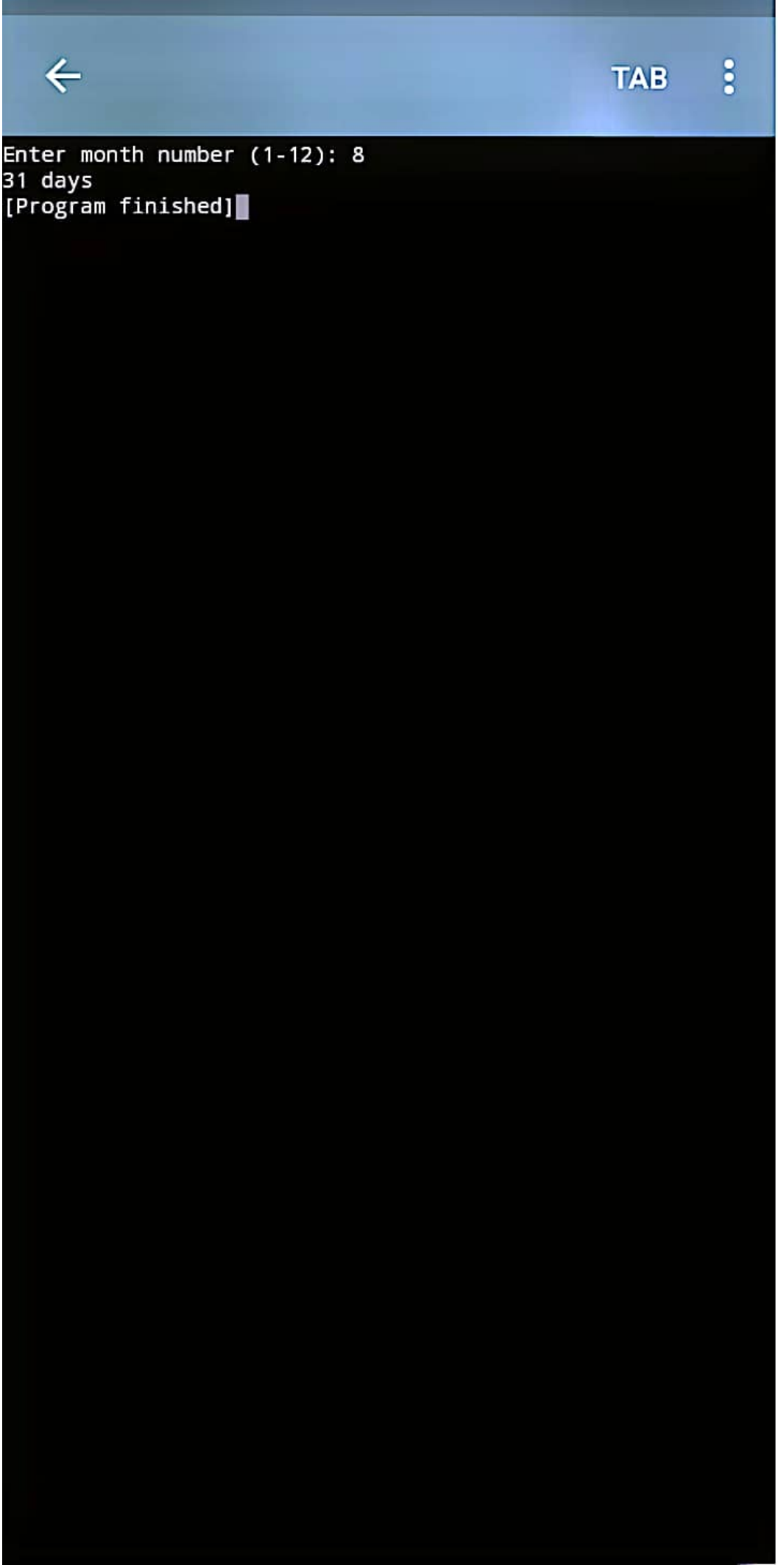
output:

```
Enter month number : 8
31 days.
```

Scanned with CamScanner



Scanned with CamScanner



Scanned with CamScanner

vowel or consonant using switch case:

```
#include <iostream>
using namespace std;
int main () {
    char ch;
    cout <<"Enter alphabet :";
  cin >> ch;
  switch (ch){
case 'a': case 'e' : case 'i': case 'o' : case 'u' :
    cout <<"vowel"; break;
  default :
  cout<< "consonant";
          }
  return 0;
    }
```

output:

```
Enter alphabet : M
consunent.
```

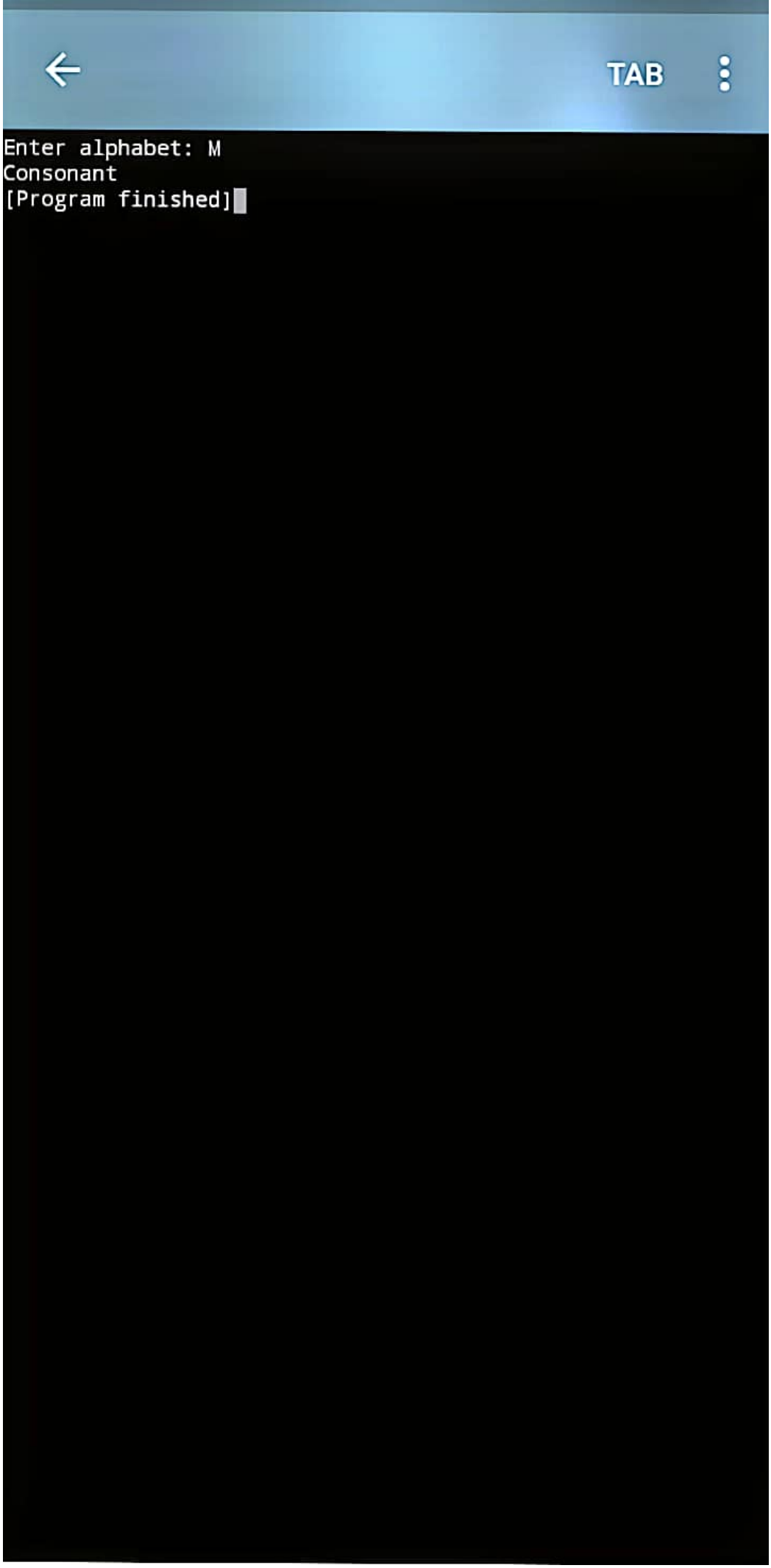
Scanned with CamScanner

new*

```
1  #include <iostream>
2  using namespace std;
3  int main() {
4     char ch;
5     cout << "Enter alphabet: ";
6     cin >> ch;
7     switch(ch){
8        case 'a': case 'e': case 'i': case 'o':
   case 'u':
9        case 'A': case 'E': case 'I': case 'O':
   case 'U':
10          cout << "Vowel"; break;
11       default:
12          cout << "Consonant";
13    }
14    return 0;
15 }
```

Tab | { | } | : | ; | "

Scanned with CamScanner



Scanned with CamScanner